4.3 Greatest Common Divisor and the Euclidean Algorithm

Monday, October 14, 2024 8:35 AM

Definition: An integer $p > 1$ is called prime iff p only has positive factors 1 and p.

The Fundamental Theorem of Arithmetic

Every integer greater than 1 can be written uniquely as a prime, or product of at least two primes.

$100 \stackrel{?}{=} 2 \cdot 50 = 2 \cdot 2 \cdot 25 = 2^2 \cdot 5^2$ $1024 \stackrel{?}{=} 2 \cdot 512 = 2 \cdot 2 \cdot 256$

$75 \stackrel{?}{=} 5 \cdot 15 = 3 \cdot 5^2$ $= 2 \cdot 2 \cdot 2 \cdot 128 = 2 \cdot 2 \cdot 2 \cdot 2 \cdot 64$

These are prime factorizations! $= \cdots = 2^{10}$

Def: If an integer is not prime it is composite.

Theorem: If n is composite, it has a prime divisor less than or equal to $\sqrt{n}$.

This means, to check prime-ness we need to only look up to $\sqrt{n}$.

Why does this work?

Trial division is an algorithm for testing primeness by checking divisibility with every prime between 2 & $\sqrt{n}$.

Show 101 is prime!

$\sqrt{101}$ is a little more than 10, but not 11. So we need only check 2, 3, 5, 7 for divisibility.

All of these leave a remainder, so 101 is prime.

Theorem: There are infinitely many primes.

Def: Let $a,b \in \mathbb{Z}$ with at least one non-zero. Then the largest integer d satisfying $d|a$ and $d|b$ is the greatest common divisor of a & b.

This is denoted $\gcd(a,b)$ or (a,b).

What is $\gcd(30,64)$?

$30 = 2 \cdot 3 \cdot 5 \qquad 64 = 2 \cdot 2 \cdot 2 \cdot 2 \cdot 2 \cdot 2$

so $\gcd(30,64) = 2$.

Def: Integers a and b are relatively prime iff $\gcd(a,b) = 1$.

Def: If $a,b \in \mathbb{Z}$ and m is the smallest integer satisfying $a|m$ and $b|m$, then m is the least common multiple of a & b.

Denoted by $\operatorname{lcm}(a,b)$.

What is $\operatorname{lcm}(12,30)$?

$12 = \underline{2}^2 \cdot 3 \qquad 30 = 2 \cdot \underline{3} \cdot \underline{5}$

$$\operatorname{lcm}(12,30) = 2^2 \cdot 3 \cdot 5 = \boxed{60}$$

If $\{p_1, p_2, \ldots, p_n\}$ is the set of all prime factors of a and b,

$$\gcd(a,b) = p_1^{min} \cdot p_2^{min} \cdots p_n^{min}$$ (min is the lowest power in a or b, even 0)

$$\operatorname{lcm}(a,b) = p_1^{max} \cdot p_2^{max} \cdots p_n^{max}$$ (max is the highest power in a or b, at least 1)

The Euclidean Algorithm

We use the algorithm to find the greatest common divisor of two integers.

Allows us to avoid prime factorization

which can be hard for large integers.

Consider $\gcd(12, 44) = x$

$$44 = 3(12) + 8$$

If x divides 44 and 12, then

x divides $44 - 3(12) = 8$.

Then x is also the $\gcd(8, 12)$.

$$12 = 1 \cdot (8) + 4$$

If x divides 12 and 8, then

x divides $12 - 1 \cdot (8) = 4$.

Then x is also the $\gcd(4, 8)$.

$$8 = 2(4).$$

Since $4 \mid 8$, $\gcd(4,8) = 4$

and $\gcd(12,44) = \gcd(8,12) = \gcd(4,8) = 4$.

<u>Lemma</u>: Let $a, b, q, r \in \mathbb{Z}$, then

if $a = qb + r$, $\gcd(a,b) = \gcd(b,r)$

<u>The Algorithm</u>

(Let $a, b \in \mathbb{Z}$, $a > b$)

initialize $x := a$

initialize $y := b$

while $(y \neq 0)$:

- $r := x \bmod y$
- $x := y$
- $y := r$

return x ($x = \gcd(a,b)$)

Find $\gcd(12, 18)$.

$$r_1 = \underset{x}{18} \bmod \underset{y}{12} = \underset{r}{6}$$

$x = 12$

$y = 6$

$$r_2 = \underset{x}{12} \bmod \underset{y}{6} = \underset{r}{0}$$

$x = 6$
$y = 0$! $\boxed{\text{return } 6}$

$\gcd(12, 18) = 6$

Find $\gcd(21, 34)$ using Euclidean Algorith

$r_0 = 34 \bmod 21 = 13$

$r_1 = 21 \bmod 13 = 8$

$r_2 = 13 \bmod 8 = 5$

$r_3 = 8 \bmod 5 = 3$

$r_4 = 5 \bmod 3 = 2$

$r_5 = 3 \bmod 2 = 1$

$r_6 = 2 \bmod 1 = 0$

$\boxed{\gcd(21, 34) = 1}$

Find the $\gcd(34, 54)$ using Euclidean

$r_0 = 54 \bmod 34 = 20$

$r_1 = 34 \bmod 20 = 14$

$r_2 = 20 \bmod 14 = 6$

$r_3 = 14 \bmod 6 = 2$

$r_4 = 6 \bmod 2 = 0$

$\boxed{\gcd(34, 54) = 2}$